



AI-POWERED DRIVER WELLNESS & SAFETY MONITORING SYSTEM

Final Project Presentation

From problem definition → data → architecture → training → evaluation → integration → real-time risk fusion → deployment

5 ML modules • 5 team members • 1 integrated inference pipeline • 1 continuous 0–100 wellness score

Problem Statement

Why driver monitoring must move beyond a single fatigue flag

THE SAFETY GAP

- Driver risk is multi-causal: fatigue, distraction, phone use, smoking, drinking and seat-belt non-compliance can occur simultaneously.
- A single model or a single frame cannot provide a reliable picture of the driver's current state.
- Fleet and in-cabin monitoring needs a continuous, objective signal rather than isolated predictions.
- A single cabin camera can expose multiple safety signals if lightweight models can run together.

PROJECT VISION

ONE CAMERA



FIVE INDEPENDENT SAFETY SIGNALS



ONE FUSED WELLNESS SCORE

A bounded 0–100 score that reflects concurrent risk instead of forcing the system to choose one cause.

Objectives

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Engineering and research goals across the six milestones

Detect

fatigue and unsafe in-cabin behaviours

Differentiate

temporal, landmark, activity and object-based evidence

Standardize

module outputs through common adapters

Fuse

heterogeneous risk signals into one score

Stabilize

predictions using temporal logic where needed

Deploy

a usable Gradio application on the target platform

Milestone Journey

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How the project progressed from idea to deployable prototype

M1 Problem + literature

Define safety problem, scope, stakeholders, initial solution paradigm

M2 Dataset preparation

Collect, clean, relabel, split and validate task-specific datasets

M3 Architecture design

Compare candidate models; define I/O, loss, metrics and compute budget

M4 Training + tuning

Train all five modules; tune hyperparameters; select best checkpoints

M5 Evaluation + analysis

Freeze models; evaluate; analyze errors; prepare integration evidence

M6 Integration + deployment

Adapters, streaming, risk fusion, Gradio, end-to-end validation, deployment

Milestone 1 — Problem Definition & Motivation

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Establishing the system's safety scope

WHAT M1 ESTABLISHED

- Continuous driver wellness monitoring as an additional safety layer.
- Dashboard / embedded cabin camera as the primary hardware scope.
- Stakeholders: logistics agencies, cab owners, bus operators and individual drivers.
- Out of scope: mechanical braking integration, native mobile application and cloud fleet dashboard.
- Initial research direction: combine facial cues, behavior detection and temporal reasoning.

INITIAL SAFETY SIGNALS

Eye closure / PERCLOS

Yawning

Head pose

Phone / distraction

Smoking / drinking

Seat-belt compliance

M1 outcome: a clear multi-signal problem definition that could be decomposed into five independently trainable modules.

Milestone 2 — Dataset Preparation

Building task-specific datasets with leakage control

Common principle

Clean → label → verify → split → preprocess → document

Module	Dataset / Source	Core preparation	Output
Video Fatigue	UTA-RLDD	Sequence construction; subject-disjoint split	16-frame clips
Landmark Fatigue	YawDD	MediaPipe features; 45-frame windows	EAR/MAR/Pose sequences
Driver Activity	AUC Distracted Driver	Stratified image split; augmentation	5 activity classes
Seatbelt / Phone	DMS / YOLO dataset	Annotation verification; YOLO format	Phone + Seatbelt boxes
Smoking / Drinking	Roboflow merged set	Cleaning; duplicate removal; class balancing	Smoking + Drinking boxes

Key M2 discipline: temporal windows stay within their native video and subject-level splitting is used where applicable to reduce leakage.

Milestone 3 — Model Architecture Design

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Five architectures selected for complementary evidence + real-time feasibility

Video Fatigue

CNN-LSTM → later EfficientNet-B0 + BiLSTM

Temporal visual cues

Landmark Fatigue

LSTM

EAR / MAR / head pose

Driver Activity

MobileNetV3

Lightweight frame classification

Seatbelt / Phone

YOLOv8n

Fast explainable localization

Smoking / Drinking

YOLOv8n

Real-time object detection

Design principle: the lightweight model wins unless the accuracy gap is large enough to justify additional latency and memory.

Milestone 4 — Training, Tuning & Checkpointing

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Turning architecture specifications into deployable model artifacts

TRAIN

Task-specific training on cleaned datasets

TUNE

Learning rate, batch size, optimizer, temporal settings, dropout etc.

EVALUATE

Accuracy / F1 for classification; mAP / precision / recall for detection

SELECT

Best checkpoint based on validation evidence

Common experimental environment: Python • PyTorch / Ultralytics • OpenCV • MediaPipe • scikit-learn • NVIDIA T4 / Google Colab

M4 deliverable: trained checkpoints and inference pipelines ready for the integration stage.

Milestone 5 — Evaluation & Error Analysis

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Freezing the models and understanding where they succeed or fail

EVALUATION

- Freeze selected checkpoints and evaluate on held-out validation/test data.
- Use task-appropriate metrics: accuracy, precision, recall, F1 and confusion matrices for classification.
- Use mAP@50 and mAP@50–95 for object detection.
- Measure inference latency and practical deployment envelope.
- Analyze per-class failures rather than relying only on aggregate scores.

KEY LESSONS

- Video fatigue showed an unseen-subject generalization gap.
- Landmark fatigue is camera/face-visibility sensitive.
- Driver Activity maintained strong accuracy and speed.
- Smoking/Drinking: smoking is stronger than drinking; small objects remain difficult.
- Seatbelt/Phone: strong documented detection performance; real-world glare/shadow remain important.

Five Modules — Final System Roles

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Each member owns one complementary source of evidence

Module	Member	Architecture	Primary evidence	Final role
Video Fatigue	Kushagra	EfficientNet-B0 + BiLSTM	Temporal visual fatigue	Complementary fatigue signal
Landmark Fatigue	Shiwani	MediaPipe + LSTM	EAR / MAR / pose	Interpretable fatigue signal
Driver Activity	Shubham	MobileNetV3-Large	Driving behaviour	Distraction / phone activity
Smoking & Drinking	Ravina	YOLOv8n	Unsafe objects/actions	Smoking / drinking events
Seatbelt & Phone	Sohini	YOLOv8n	Compliance / phone	Seatbelt + phone events

The modules are deliberately heterogeneous: different evidence sources reduce dependence on a single model or failure mode.

Final Module Performance Snapshot

Reported results are task-specific and therefore not directly comparable

32.86%

Video Fatigue test accuracy

0.831

Landmark confidence in M6
integration test

93.14%

Driver Activity test accuracy

0.8197

Smoking/Drinking mAP@50

0.9526

Seatbelt/Phone mAP@50

Performance interpretation

- Driver Activity and YOLO-based behaviour modules provide the strongest documented quantitative performance.
- Fatigue models are more sensitive to subject, camera and temporal-domain shift.
- Metrics describe individual tasks; they should not be averaged into one 'model accuracy'.

M6 Risk Fusion: From Individual Predictions to Unified Driver Risk

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Milestone 6 introduced the final risk-scoring layer that converts five independent model outputs into one interpretable Driver Wellness Score.

BEFORE M6: INDEPENDENT OUTPUTS

Video Fatigue

Landmark Fatigue

Driver Activity

Smoking / Drinking

Seatbelt / Phone

Each module produces its own prediction and confidence.

M6 RISK FUSION ENGINE

PredictionResult

Severity × Confidence

Individual Event Risk

Total Risk = $\sum R_i$

0–100 Wellness Score

WHAT M6 IMPROVED

1 Multi-cause risk

Multiple unsafe behaviours can contribute simultaneously.

2 Confidence-aware

Confidence scales the contribution; it does not create risk by itself.

3 Failure-aware

Unavailable / failed modules are excluded, not treated as safe.

4 Stable output

A 3-second moving average reduces frame-to-frame flicker.

M6 transforms five independent ML predictions into one continuous, interpretable and temporally stable driver-safety signal.

Risk Fusion — Step 1: Event Risk

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Every available module contributes only through a standardized PredictionResult

1 • EVENT RISK

$$R_i = \text{severity}_i \times \text{confidence}_i$$

Severity = how dangerous the event is (0–10)

Confidence = how certain the model is (0–1)

Risk = danger × certainty

M6 SEVERITY EXAMPLES

Event	Severity
Texting phone	9
Talking phone	8
Drowsy / yawning	7
Drinking	10
Smoking	5
No seatbelt / no detection	8

Important: confidence is clamped to [0,1], so an overconfident/malformed module cannot inject an invalid risk value.

Risk Fusion — Step 2 & 3: Safe Events and Total Risk

Only genuine risk accumulates—and failed modules are not treated as 'safe'

SAFE EVENT RULE

- Safe predictions are explicitly mapped to $R_i = 0$.
- Examples: safe driving, alert/normal fatigue state, no smoking/drinking, seatbelt-only state.
- Confidence is ignored for safe labels because they are defined as non-risk events.

Safe $\neq$ uncertain

A valid safe result contributes zero.

TOTAL RISK

$$R_{\text{total}} = \sum R_i$$

- Sum across AVAILABLE modules only.
- Warm-up, crashed or unavailable modules are excluded—not silently assigned zero risk.
- This prevents a failed detector from making the driver appear safer.
- All adapters therefore share the same status-aware contract.

Risk Fusion — Step 4: Exponential Wellness Score

Turning accumulated risk into a bounded 0–100 signal

CORE FORMULA

$$\text{Score} = 100 \times (1 - e^{(-k \cdot R_{\text{total}})})$$

$$k = 0.05$$

- Bounded: the score naturally remains within 0–100.
- Additive: concurrent risks compound rather than being diluted by safe modules.
- Saturating: once risk is already very high, additional events have a smaller incremental effect.

Risk Fusion — Step 5 & 6: Smooth and Classify

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Making the live score stable and human-readable

TEMPORAL SMOOTHING

**Displayed score = mean(raw scores
over last 3 seconds)**

Why?

- Single-frame false positives should not cause the gauge to jump.
- Short detector dropouts should not immediately erase sustained risk.
- Temporal averaging reflects behavior over a short, human-meaningful interval.

RISK LEVELS

0–25

LOW RISK

25–50

MODERATE RISK

50–75

HIGH RISK

75–100

CRITICAL RISK

Risk Fusion — Worked Example

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Drowsy + texting + unbelted: how the final score is produced

Module	Prediction	Severity	Confidence	Event Risk
Landmark	Drowsy	7	0.90	6.3
Driver Activity	texting_phone	9	0.80	7.2
Seatbelt	No Detection	8	0.75	6.0
Video Fatigue	Low / Safe	0	—	0
Smoking	none / Safe	0	—	0

CALCULATION

$$R_{\text{total}} = 6.3 + 7.2 + 6.0 \\ = 19.5$$

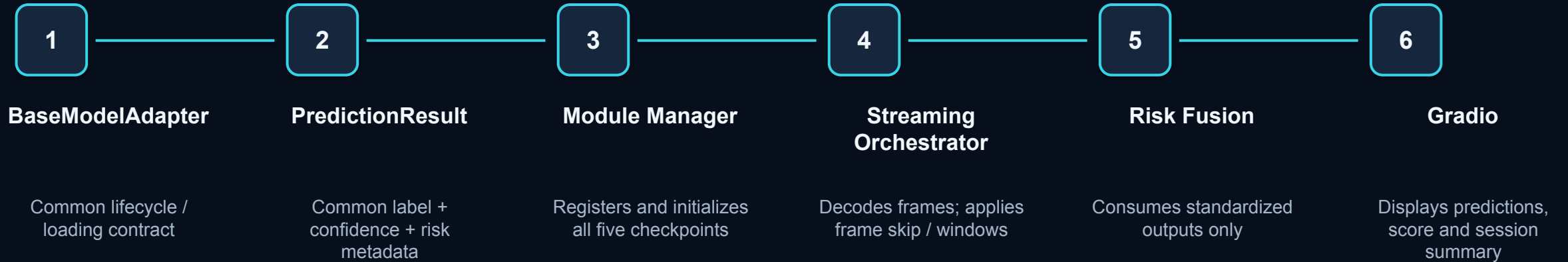
$$\text{Score} = 100 \times (1 - \\ e^{(-0.05 \times 19.5)}) \\ = 62.29$$

HIGH RISK

Two safe modules contribute nothing; three active risks compound.

Individual Module → Integrated System Transition

How five independent implementations became one pipeline



Key engineering idea: each module remains independently maintainable, while the integration layer knows only the standardized contract—not the internal model architecture.

End-to-End Integrated Pipeline

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One video stream, five adapters, one risk decision



STREAMING BEHAVIOUR

Frame decode → module-specific frame sampling → temporal windows / smoothing → standardized result → risk fusion → UI update

M6 integration test: 1,760 frames • all five modules active • no module errors • 49.8/100 Moderate Risk

End-to-End Validation

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Demonstrating that the modules actually work together

1,760

Frames processed

106.7 s

Elapsed time

5 / 5

Modules active

0

Module errors

49.8 / 100

Observed wellness score

OBSERVED RESULT

MODERATE RISK

Landmark prediction: Drowsy
Confidence: 0.831

IMPORTANT INTERPRETATION

This is an integration / pipeline validation result—not a system-wide accuracy benchmark.

The test confirms that all five modules can contribute to the common risk score.

Robustness & Failure Handling

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The integrated system fails safely and reports module state explicitly

Unavailable face

Landmark module reports a structured unavailable/error state.

Short video

Insufficient temporal input is handled gracefully.

Model warm-up / failure

Unavailable modules are excluded from fusion rather than treated as safe.

YOLO flicker

Confidence filtering, NMS and temporal stabilization reduce transient detections.

Streaming load

Frame skipping and rolling histories reduce redundant computation.

Gradio Interface

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Turning the ML pipeline into a usable driver-monitoring application

RECORDED VIDEO

Upload driver video

Analyze

Annotated output

Module predictions

Wellness gauge

Session summary

LIVE WEBCAM

Start webcam

Streaming inference

Live annotations

Real-time score

Risk-level updates

Stop + summary

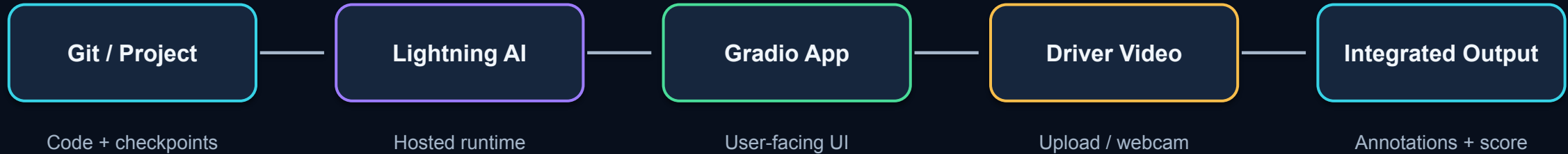
Cockpit-style dashboard • module status chips • risk gauge • visual hierarchy

Deployment — Lightning AI + Gradio

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Packaging the integrated application for hosted inference

FINAL DEPLOYMENT FLOW



DEPLOYMENT ENGINEERING

- Thin application wrapper + centralized inference core.
- Model caching / lazy loading and frame skipping reduce startup and runtime resource pressure.
- requirements.txt and README.md document the reproducible runtime.
- Gradio exposes the same integrated inference pipeline rather than separate module demos.

Platform note: the consolidated M6 documentation records a hosted Space deployment with runtime quota constraints; this presentation reflects the team's final Lightning AI hosting update.

Deployment Challenges & Mitigations

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What had to be considered beyond model accuracy

Challenge	Impact	Mitigation / direction
GPU / runtime quota	Inference may be interrupted or unavailable	Caching, lazy loading, lighter models, dedicated GPU option
Five-model memory	Simultaneous checkpoints increase VRAM demand	Lazy initialization; optimize model lifecycle
Long videos	GPU execution time can grow quickly	Frame skipping; shorter windows; future chunking
Landmark throughput	Face-landmark extraction can bottleneck webcam FPS	Optimize MediaPipe / asynchronous processing
Domain shift	Cabin lighting, glare, camera angle affect performance	Broader real-world validation

Deployment readiness is therefore an engineering problem as well as a machine-learning problem.

Individual Contributions — Member Ownership

Final contributions across the project lifecycle

Member	Module	Key technical ownership	Integration contribution
Kushagra	Video Fatigue	Sequence modeling; EfficientNet-B0 + BiLSTM; fatigue evaluation	Checkpoint + standardized fatigue output
Shiwani	Landmark Fatigue	MediaPipe features; LSTM; temporal fatigue logic	Landmark adapter; temporal state + failure handling
Shubham	Driver Activity	MobileNetV3-Large; activity evaluation	DriverActivityAdapter; Gradio lead; streaming API
Ravina	Smoking & Drinking	YOLOv8n; dataset cleaning; tuning; held-out evaluation	Detection adapter + deployment-oriented consolidation
Sohini	Seatbelt & Phone	YOLOv8n; robustness; temporal consensus	Adapter; Risk Fusion integration; final documentation/consolidation

Shared responsibility: consistent interfaces, common inference flow, integration testing, documentation and final project delivery.

Project-Level Limitations

What prevents us from calling this a production safety-certified system

MODEL LIMITATIONS

- Fatigue generalization across unseen drivers remains the largest technical gap.
- Landmark fatigue depends on face visibility and camera orientation.
- Drinking detection is weaker than smoking detection.
- Small-object localization and real-world cabin conditions remain challenging.
- Metrics across modules are task-specific and not directly comparable.

SYSTEM / DEPLOYMENT LIMITATIONS

- Five simultaneous models can require significant GPU memory.
- Live webcam throughput is constrained by landmark extraction.
- Deployment runtime can be constrained by hosted GPU quota/time limits.
- More diverse real-world and longer-duration validation is needed.
- Not safety-certified; intended as a research/engineering prototype.

Future Work

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From a strong integrated prototype toward robust real-world deployment

Model

Cross-subject fatigue validation; confidence calibration; improved drinking detection

Efficiency

Quantization, pruning, distillation, asynchronous execution

Streaming

Faster landmark processing; optimized decoding; WebRTC

Deployment

Dedicated GPU-backed hosting; health monitoring; model versioning

Validation

Night/glare/occlusion; camera positions; longer live sessions; independent end-to-end ground truth

CONCLUSION

**Five models.
One pipeline.
One wellness signal.**

The project evolved from independent module research into a unified, interpretable and deployable driver-monitoring prototype.

- ✓ Five trained modules
- ✓ Standardized adapters
- ✓ Updated M6 risk fusion
- ✓ End-to-end integration
- ✓ Gradio interface
- ✓ Hosted deployment pathway

THANK YOU

Questions & Discussion